

Mobile Development

Dart Essentials - Part 1

LAB 1/W2

3ND YEAR - ENSIA

OBJECTIVES

- Get hands-on practice with Dart basics: functions, loops, and strings.
- Strengthen algorithmic thinking without relying on built-in functions.
- Build confidence solving simple problems before moving to Flutter.
- See if students can solve these exercises without the use of ChatGPT.

TASKS BEFORE THE LAB

- Install the following tools:
 - VS
 - Dart SDK
 - VS Extensions related to Dart

INDIVIDUAL EXERCISES

DON'T USE PREDEFINED FUNCTIONS

Task / Exercise C1 (Dart : Simple Functions) [60 mins]:

- Check whether a given string is a palindrome (no use of predefined functions).
- Convert a decimal number to a binary number .

Task / Exercise C2 (Dart : Loops) [60 mins]:

Implement the following functions in Dart :

- Write a function to check whether a given number is prime. You can make use of the function `Math.sqrt( DOUBLE )`
- Compute the factorial for a given number using :
 - Iteration.
 - Recursive function.

Task / Exercise C3 (Dart : Processing) [60 mins]:

- Write functions to do the following :
 - Capitalize a given text (First letter of all words is capital).
 - Remove duplicate elements in a given **Array of Strings**. (don't convert to Set)
 - Write a function that returns True when a string contains only a single substring of **00**.

SUBMISSION FORM

Students must submit their solutions using this week. Deadline for the submission is Sunday :

<https://forms.gle/sXHA4jzZ3QXSPqbZ9>

OPTIONAL EXERCISES

Task / Exercise P1 :

Write function for the following :

1. Find all palindromes inside a text. What's the complexity of your solution?
2. Write a function that checks if two strings are anagrams of each other.

Task / Exercise P2 :

Given a string containing brackets [], braces {}, parentheses (), or any combination thereof, verify that any and all pairs are matched and nested correctly. The string may also contain other characters, which for the purposes of this exercise should be ignored.

Task / Exercise P3 :

Given a list of integers, implement a function which calculates the maximum sum of n consecutive elements in the list (sum of n digits next to another that have the largest sum). n is a given number.

Task / Exercise P4 :

For a given integer, find the prime factor integers where if we multiply them, they give the input number.

Example : $36 = 2 * 2 * 3 * 3$. The prime factors are : 2,2,3,3

Task / Exercise P5 :

Write a function which spells out numbers into English. In other words, a given number is written in words. Example : 151 → One hundred and fifty one.

Task / Exercise P6 :

The soundex is a useful function to compare if two strings sound the same. Implement the function using Dart.

Task / Exercise P7 :

Inventing a catchy and small word which can be pronounced in English, can be interesting to business people trying to find a good brand name for their product or company.

- Try to write a Dart script which can tell whether a word is pronounceable in English or not

Task / Exercise P8 (Highly Recommended) :

Shuffling number in a matrix : Given a matrix of 5x5 numbered in a sequential number from 1 to 24 where the last cell is blank.

Write an algorithm to shuffle randomly the cells where you can move only the cell with a number next to its adjacent blank cell.

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	→